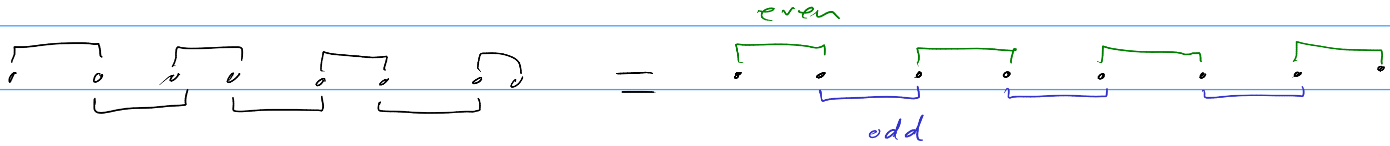
$h = -it \sum_i \gamma_i \gamma_{i+1} + g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} + \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+4}$

split hamiltonian into h_{even}, h_{odd} and rewrite majorana operators into even and odd:

$a_j := \gamma_{2n}$

$b_j := \gamma_{2n+1}$



$$h = h_{even} + h_{odd}$$

$$= -it\left(\sum_j a_j b_j + \sum_j b_j a_{j+1}\right)$$

→ Use Jordan Wigner transformation to map fermionic (majorana) operators to spin operators (Pauli matrices)

$a_j = K_j \sigma_j^x$

$b_j = K_j \sigma_j^y$

$a_j b_j = i \sigma_j^z$

$K_j := \prod_{l=1}^{j-1} \sigma_l^z$

$$-it\left(\sum_j i\sigma_j^z + K_j \sigma_j^y \underline{K_{j+1}} \sigma_{j+1}^x\right)$$

$$= -it\left(\sum_j \left(i\sigma_j^z + K_j \sigma_j^y K_j \sigma_j^z \sigma_{j+1}^x\right)\right)$$

$$-it\left(\sum_j i\sigma_j^z + \cancel{K_j^2} \underbrace{\sigma_j^y \sigma_j^z}_{= i\sigma_j^x} \sigma_{j+1}^x\right)$$

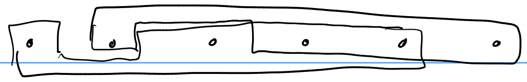
$$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$$

$$= -it\left(\sum_j i\sigma_j^z + i\sigma_j^x \sigma_{j+1}^x\right)$$

$$= \sum_j -\overset{h}{\cancel{t}} \sigma_j^z - \overset{J}{\cancel{t}} \sigma_j^x \sigma_{j+1}^x = \sum_j -J \sigma_j^x \sigma_{j+1}^x - h \sigma_j^z \quad \therefore \text{as expected.}$$

let $t = J = h$

$$g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} = g \sum_j \underline{a_j a_{j+1} b_{j+1} a_{j+2}} + \underline{b_j b_{j+1} a_{j+2} b_{j+2}}$$

 $= a_j \; b_j \; a_{j+1} \; b_{j+1} \; a_{j+2} \; b_{j+2}$

$$a_j \underline{a_{j+1} b_{j+1}} \underline{a_{j+2}} =$$

$$K_j \sigma_j^x \cdot i \sigma_j^z \cdot K_{j+2} \sigma_{j+2}^x = \cancel{K_j} \sigma_j^x \, i \sigma_{j+1}^z \cdot \cancel{K_j} \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$= \sigma_j^x \, i \, \sigma_{j+1}^z \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$i \sigma_j^x \sigma_j^z \underbrace{\cancel{\sigma_{j+1}^z \sigma_{j+1}^z}}_{1} \sigma_{j+2}^x$$

$$i(-i \sigma_j^y) \sigma_{j+2}^x = \sigma_j^y \sigma_{j+2}^x$$

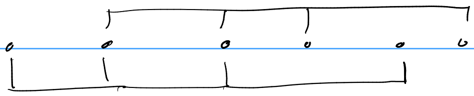
$$b_j \underline{b_{j+1}} \underline{a_{j+2} b_{j+2}} = \cancel{K_j} \sigma_j^y \cdot \cancel{K_j} \sigma_j^z \sigma_{j+1}^y \cdot i \sigma_{j+2}^z$$

$$\sigma_j^y \sigma_j^z \sigma_{j+1}^y \, i \sigma_{j+2}^z$$

$$\cancel{i}\, i \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z = -\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$g \sum \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} = g \sum \sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$+\gamma_i\gamma_{i+1}\gamma_{i+2}\gamma_{i+3} = \sum_j \underline{a_j b_j a_{j+1} a_{j+2}} + \underline{b_j a_{j+1} b_{j+1} b_{j+2}}$$



$= \quad a_j \quad b_j \quad a_{j+1} \quad b_{j+1} \quad a_{j+2} \quad b_{j+2}$

$$\cancel{a_j b_j} \cdot a_{j+1} \cdot a_{j+2} = i\sigma_j^z \cdot \cancel{K_j}\sigma_j^z \sigma_{j+1}^x \cdot \cancel{K_j}\sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$= i\sigma_j^z \cancel{\sigma_j^z \sigma_j^z} \sigma_{j+1}^x \sigma_{j+1}^z \sigma_{j+2}^x$$

xy zx $i\sigma_{j+1}^x$

$$i\sigma_j^z(-i\sigma_{j+1}^y)\sigma_{j+2}^x = \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x$$

$$b_j a_{j+1} b_{j+1} b_{j+2} = \cancel{K_j}\sigma_j^y \cdot i\sigma_{j+1}^z \cdot \cancel{K_j}\sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^y$$

$$= \sigma_j^y \sigma_j^z (i\,\underbrace{\cancel{\sigma_{j+1}^z \sigma_{j+1}^z}}_{1})\sigma_{j+2}^y$$

$$= (i\sigma_j^x)(i)(\sigma_{j+2}^y) = -\sigma_j^x \sigma_{j+2}^y$$

$$g\sum \gamma_i\gamma_{i+1}\gamma_{i+2}\gamma_{i+3} = g\sum \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y$$

→ flipped sign of second term and reversed x, y, z index order
↳ makes sense.

$$H = -t\sum_j(\sigma_j^z + \sigma_j^x\sigma_{j+1}^x) + g\sum(\sigma_j^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z) + (\sigma_j^z\sigma_{j+1}^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+2}^y)$$

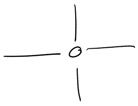
$$H = -t\sum_j(\sigma_j^z + \sigma_j^x\sigma_{j+1}^x) + g\sum_j(\sigma_j^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+2}^y) - (\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z - \sigma_j^z\sigma_{j+1}^y\sigma_{j+2}^x)$$

To study ferromagnetic phases:

$$H = \sum_j J\sigma_j^x\sigma_{j+1}^x + h\sigma_j^z - g(\sigma_j^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+2}^y) - g(\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z - \sigma_j^z\sigma_{j+1}^y\sigma_{j+2}^x)$$

MPO: Check first term:

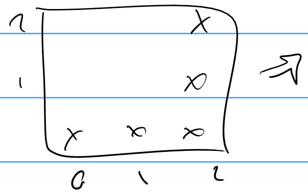
$$h = J \sum_j \sigma_j^x \sigma_{j+1}^x + \sigma_j^z$$



σ^x	σ^x	I ✓
I	σ^x	σ^x ✓
—x—	I	I ✓
I	I	—x—
I	σ^z	I ✓

$$\begin{array}{c|cc} 1\ 0 \to & \sigma^z & 0 \\ \sigma\ 1 \to & \sigma^x & \sigma^x \\ -x- 2 \to & I & 0 \\ \hline & I & \sigma & -x- \\ & 0 & 1 & 2 \end{array} \Longrightarrow \begin{array}{c|ccc} -x- & I & 0 & 0 \\ \sigma^x & \sigma^x & 0 & 0 \\ I & \sigma^z & \sigma^x & I \\ \hline & I & \sigma^x & -x- \end{array}$$

$$\begin{pmatrix} I & 0 & 0 \\ \sigma^x & 0 & 0 \\ \sigma^z & \sigma^x & I \end{pmatrix} \to \begin{pmatrix} 0 & 0 & I \\ 0 & 0 & \sigma^x \\ I & \sigma^x & \sigma^z \end{pmatrix}$$



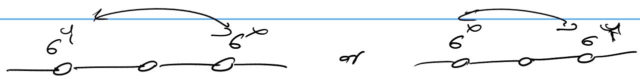
1

σ^z

$1 \quad \sigma^z \quad \sigma^x$

$\sigma^z\sigma^z \quad \sigma^x$ ✓

$$g(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y)$$



or

①

$\sigma^y I$	σ^x	I
σ^y	I	σ^x
I	σ^y	$I\sigma^x$

$I \quad I \quad$ —x—

—x— $\quad I \quad I$

②

$-\sigma^x I$	σ^y	I
$-\sigma^x$	I	σ^y
I	$-\sigma^x$	$I\sigma^y$

$$\begin{array}{c|cccccc} \cancel{-x-} & 1 & 0 & 0 & 0 & 0 & 0 \\ \sigma^y I & -\sigma^x & 0 & 0 & 0 & 0 & 0 \\ \sigma^x I & \sigma^y & 0 & 0 & 0 & 0 & 0 \\ \sigma^y & 0 & -1 & 0 & 0 & 0 & 0 \\ \sigma^x & 0 & 0 & 1 & 0 & 0 & 0 \\ I & 0 & 0 & 0 & -\sigma^y & \sigma^x & 1 \\ \hline & 1 & \sigma^x & \sigma^y & \sigma^x I & \sigma^y I & \cancel{-x-} \end{array}$$

$$g(\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x)$$

I	I	—x—
1	σ^x	$\sigma^y\sigma^z$
σ^x	σ^y	σ^z
$\sigma^x\sigma^y$	σ^z	1
—x—	I	1

I	$-\sigma^z$	$\sigma^y\sigma^x$
σ^z	σ^y	σ^x
$\sigma^z\sigma^y$	σ^x	1

1

σ^x

σ^z

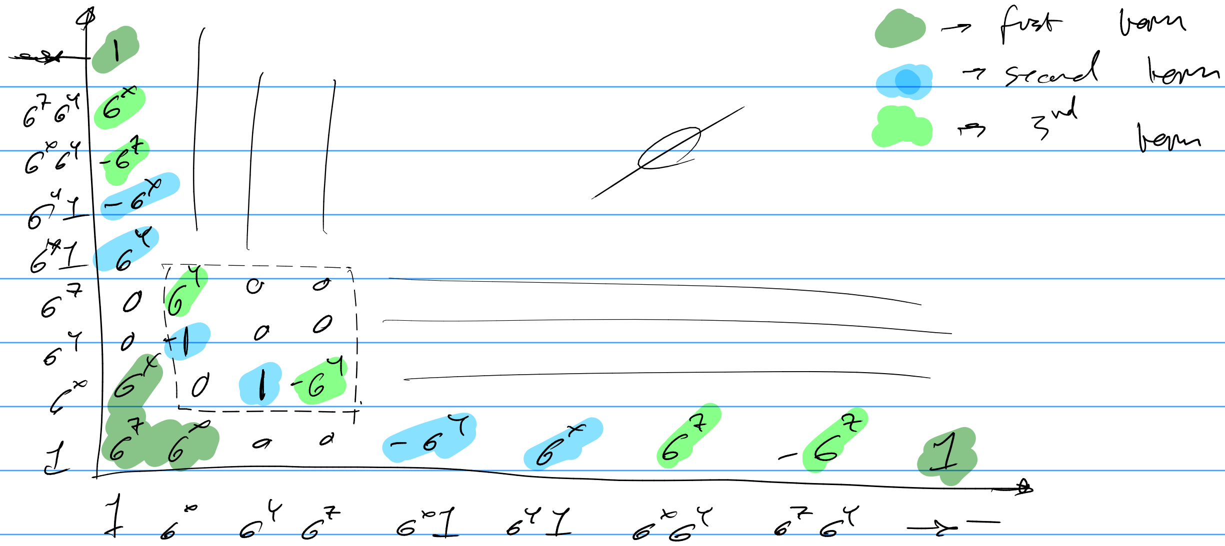
$\sigma^x\sigma^y$

$\sigma^z\sigma^y$

—x—

$$\begin{array}{c|cccccc} -x- & 1 & 0 & 0 & & & 0 \\ \sigma^z\sigma^y & \sigma^x & 0 & 0 & 0 & 0 & 0 \\ \sigma^x\sigma^y & -\sigma^z & 0 & 0 & 0 & 0 & 0 \\ \sigma^z & 0 & \sigma^y & 0 & 0 & 0 & 0 \\ \sigma^x & 0 & 0 & -\sigma^y & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & \sigma^z & -\sigma^x & 1 \\ \hline & 1 & \sigma^x & \sigma^z & \sigma^x\sigma^y & \sigma^z\sigma^y & \cancel{-x-} \end{array}$$

Combine all matrices together:



Try again

$$h = J\sigma_j^x \sigma_{j+1}^x + h\sigma_j^z - k\left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y + \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x\right)$$

$H = \frac{J}{2} S^+ S^- + \frac{J}{2} S^- S^+ + J^z S^z S^z - h S^z$

$I \xrightarrow{I} I$

S^+_-

S^z

$-x-$

(1) → ?

I	$-hS^z$	I
I	$\frac{J}{2} S^-$	S^+
I	$\frac{J}{2} S^+$	S^-
I	$J^z S^z$	S^z
I	I	$-x-$

(2) →

$S^z \to (1)$ ×

$S^+ \to S^+_-(2)$ ×

$S^+ \to S^-$ ×

$S^+ \to S^z$ ×

$S^z \to -x-$ ✓

$S^z(\frac{J}{2})$

$|I\rangle$

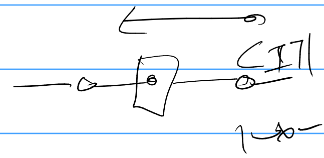
$|S^+\rangle$

$|S^-\rangle$

$|S^z\rangle$

$|x\rangle$

$$
\begin{array}{c|ccccc}
 & |I\rangle & |S^+\rangle & |S^-\rangle & |S^z\rangle & |-x-\rangle \\
\hline
|I\rangle & I & 0 & 0 & 0 & 0 \\
|S^+\rangle & S^+_- & 0 & 0 & 0 & 0 \\
|S^-\rangle & S^- & 0 & 0 & 0 & 0 \\
|S^z\rangle & S^z & 0 & 0 & 0 & 1 \\
|-x-\rangle & -hS^z & \frac{J}{2} S^- & \frac{J}{2} S^+ & S^z & I
\end{array}
$$



$|I\rangle$
$|S^z\rangle$
$|S^z I\rangle$
$|{-x-}\rangle$

	$\|I\rangle$	$\|S^z\rangle$	$\|S^z I\rangle$	$\|{-x-}\rangle$
$\|I\rangle$	I	0	0	0
$\|S^z\rangle$	S^z	0	0	0
$\|S^z I\rangle$	0	I	0	0
$\|{-x-}\rangle$	0	$\xi_1 S^z$	$\xi_2 S^z$	I

cannot keep $S^z I$ by itself, must finish.

$$H_{\xi_1,\xi_2} = \xi_1 \vec{S}_j \vec{S}_{j+1} + \xi_2 \vec{S}_j \vec{S}_{j+2}$$

$$= \frac{1}{2}\xi_1 \left(S^+ S^- + S^- S^+\right) + \xi_1 S^z S^z +$$

$$\frac{1}{2}\xi_2 \left(S^+_1 S^-_{1+2} + S^-_1 S^+_{1+2}\right) + \xi_2 S^z_1 S^z_{1+2}$$

	$\|I\rangle$	$\|S^+\rangle$	$\|S^-\rangle$	$\|S^z\rangle$	$\|S^+ I\rangle$	$\|S^- I\rangle$	$\|S^z I\rangle$	$\|{-x-}\rangle$
	$\|I\rangle$	0	0	0	0			
	S^+	0	0	0	0			
	S^-	0	0	0	0			
	S^z	0	0	0	0			
	0	I	0	0	0			
	0	0	I	0	0			
	0	0	0	I	0			
$\|{-x-}\rangle$	0	$\frac{1}{2}\xi_1 S^-$	$\frac{1}{2}\xi_1 S^+$	$\xi_1 S^z$	$\frac{1}{2}\xi_2 S^-$	$\frac{1}{2}\xi_2 S^+$	$\xi_2 S^z$	I

$$h = J\sigma^x_j \sigma^x_{j+1} + h\sigma^z_j - k\left(\sigma^y_j \sigma^x_{j+2} - \sigma^x_j \sigma^y_{j+2} + \sigma^x_j \sigma^y_{j+1} \sigma^z_{j+2} - \sigma^z_j \sigma^y_{j+1} \sigma^x_{j+2}\right)$$

$$\begin{array}{ccccccccc}
|I\rangle & |\sigma^x\rangle & |\sigma^y\rangle & |\sigma^z\rangle & |\sigma^x I\rangle & |\sigma^y I\rangle & |\sigma^x\sigma^y\rangle & |\sigma^z\sigma^y\rangle & |\to\rangle \\
\end{array}$$

$$\begin{pmatrix}
I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^x & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^y & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^z & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & I & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \sigma^y & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \sigma^y & 0 & 0 & 0 & 0 & 0 \\
h\sigma^z & J\sigma^x & 0 & 0 & k\sigma^y & -k\sigma^x & -k\sigma^z & k\sigma^x & I
\end{pmatrix}$$

$$\begin{array}{ccccccccc}
|I\rangle & |\sigma^x\rangle & |\sigma^y\rangle & |\sigma^z\rangle & |\sigma^x I\rangle & |\sigma^y I\rangle & |\sigma^x\sigma^y\rangle & |\sigma^z\sigma^y\rangle & |\to\rangle \\
\end{array}$$

$$\begin{pmatrix}
I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^x & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^y & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\sigma^z & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & I & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \sigma^y & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \sigma^y & 0 & 0 & 0 & 0 & 0 \\
h\sigma^z & J\sigma^x & 0 & 0 & k\sigma^y & -k\sigma^x & -k\sigma^z & k\sigma^x & I
\end{pmatrix}$$